

14. Three Address Code

```
#include <stdio.h>

#include <string.h>

int main()
{
    char expr[100];
    char temp[10];
    int i, t = 1;

    printf("Enter expression: ");
    scanf("%s", expr);

    printf("\nThree Address Code:\n");

    for (i = 0; expr[i] != '\0'; i++)
    {
        if (expr[i] == '*' || expr[i] == '/')
        {
            sprintf(temp, "t%d", t++);

            printf("%s = %c %c %c\n",
                temp,
                expr[i - 1],
                expr[i],
                expr[i + 1]);
```

```
        expr[i + 1] = temp[1];
        expr[i - 1] = '0';
    }
}

for (i = 0; expr[i] != '\0'; i++)
{
    if (expr[i] == '+' || expr[i] == '-')
    {
        sprintf(temp, "t%d", t++);

        printf("%s = %c %c %c\n",
            temp,
            expr[i - 1],
            expr[i],
            expr[i + 1]);
    }
}

return 0;
}
```

OUTPUT

```
Enter expression: a+b*c
```

```
Three Address Code:
```

```
t1 = b * c
```

```
t2 = a + t1
```

```
-----  
Process exited after 19.79 seconds with return value 0
```

```
Press any key to continue . . .
```